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The effect of laser conditioning on one printed composite and three CAD-CAM ceramic materials: microtensile bond strength after luting with a universal adhesive and hybrid resin composite

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Clinique universitaire de médecine dentaire
Division de cariology et d'endodontie

Thèse préparée sous la direction
du Professeur Julian Leprince et PD Dr Tiziana Bortolotto

Effet du conditionnement au laser sur un composite
imprimé et trois céramiques CAD-CAM : résistance à la
microtraction après collage avec un adhésif universel et
un composite hybride à base de résine.

“ The effect of laser conditioning on one printed composite and three CAD-CAM
ceramic materials: microtensile bond strength after luting with a universal adhesive
and hybrid resin composite “

Thèse
présentée à la Faculté de Médecine
de l'Université de Genève
pour obtenir le grade de Docteur en médecine
par

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de

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PARTIE FRANÇAISE

INTRODUCTION

La carie dentaire est considérée comme l'une des maladies les plus anciennes et les plus courantes chez l'homme. Il s'agit d'une maladie infectieuse chronique prévalente résultant de bactéries cariogènes adhérentes à la dent qui métabolisent les sucres pour produire de l'acide qui, avec le temps, déminéralise la structure de la dent (Rathee M. et al. 2024) (1). Les nouveaux concepts de dentisterie opératoire proposés par Fusayama (Fusayama T. 1980) (2) ont ouvert de nouvelles stratégies de traitement grâce à des connaissances avancées en cariologie et à l'introduction d'adhésifs dentaires. Les principes de restauration proposés par GV Black (Black GV. 1917) (3) tels que « Extension for Prevention », « Retention Form » et « Resistance Form » ont été modifiés et/ou remplacés par la « Minimal Invasive Dentistry » (Tyas MJ. 2000) (4). L'approche mini-invasive (MI) de la préparation des cavités vise essentiellement à minimiser l'ablation des substances dentaires afin de préserver une structure dentaire saine maximale (Sano H. et al. 2020) (5). Le concept de dentisterie mini-invasive en cariologie a évolué parce que nous avons maintenant une meilleure compréhension de la maladie. Nous pouvons intercepter son développement (Truman et al. 2002) (6) et avons les possibilités techniques d'enlever une quantité minimale de substance dentaire saine et de faire des obturations plus petites (Tyas MJ. 2000) (4).

Le développement de la science des matériaux dentaires a fourni au dentiste moderne un vaste arsenal de matériaux pour les procédures de restauration. L'utilisation de méthodes indirectes pour concevoir les restaurations a donné lieu à des procédures de fabrication permettant d'optimiser les propriétés mécaniques et

physiques des matériaux de restauration. L'avènement de céramiques et de matériaux composites usinables de haute performance, ainsi que de matériaux adhésifs et d'agents de collage améliorés, a élargi les applications potentielles des procédures de restauration indirecte dans la pratique dentaire moderne (Palin W. et al. 2005) (7).

Il est probable que les nouveaux développements de la technologie informatique ont conduit au succès de la conception et de la fabrication dentaires assistées par ordinateur (CAD/CAM) pour l'élaboration des restaurations. Des systèmes de CFAO très sophistiqués au fauteuil et en laboratoire ont été introduits dans la profession dentaire (Mantri et al. 2010) (8). La technologie CFAO utilise une méthode soustractive, c'est-à-dire le meulage à partir d'un bloc de matériau pour obtenir la forme souhaitée, et a été développée pour résoudre 3 problèmes. Le premier défi consistait à garantir une résistance adéquate de la restauration, en particulier pour les dents postérieures. Le deuxième défi consistait à créer des restaurations d'aspect naturel. Le troisième défi était de rendre la restauration des dents plus facile, plus rapide et plus précise (Davidowitz G. et al. 2011) (9). Suite au développement de la technologie CAD/CAM, de nouveaux matériaux ont été conçus par l'industrie tels que la céramique, le composite et les matériaux hybrides avec des propriétés mécaniques et esthétiques optimisées (Marchesi et al. 2021) (10).

La vitrocéramique au disilicate de lithium est une nouvelle génération de céramique thermopressive qui présente une résistance à la rupture et à la flexion supérieure à celle de la première génération de céramiques renforcées par de la leucite. Elle est composée de deux éléments : la silice, qui sert de matrice de verre, et l'oxyde de lithium (Li_2O), qui sert de flux pour diminuer la température croissante

de la matrice de verre, qui passe approximativement de 1100 °C à 2000 °C. L'application d'un agent de couplage silane après le mordantage des céramiques avec des acides à différentes concentrations est l'une des méthodes connues et recommandées pour augmenter la force de la liaison entre la céramique et le substrat dentaire. Le mordantage à l'acide fluorhydrique dissout sélectivement les surfaces contenant une matrice de verre et des cristaux. Il permet également d'augmenter l'énergie de surface pour la mise en œuvre d'un agent silane de surface acidifié. En conséquence, la liaison chimique entre la matrice inorganique de la céramique et la matrice organique des ciments résineux est améliorée (Özdemir H. et al. 2017) (11).

Une autre céramique dentaire est le polycristal d'oxyde de zirconium stabilisé à l'yttrium (Y-TZP), adopté par la technologie CAD/CAM pour les prothèses partielles fixes postérieures et antérieures. Les ébauches en zircone peuvent être traitées par fraisage et, selon le système, des ébauches partiellement ou entièrement frittées sont utilisées (Karakoca S. et al. 2009) (12). La zircone est un matériau polymorphe qui existe en trois phases différentes : monoclinique, tétragonale et cubique. La phase monoclinique est stable à basse température. La zircone pure est monoclinique à température ambiante et cette phase est stable jusqu'à 1170 °C. Au-delà de cette température, elle se transforme en phase tétragonale. À une température de 2370 °C, la zircone se transforme en une phase cubique, qui n'est stable qu'à très haute température. La forme tétragonale peut être stabilisée à température ambiante par l'ajout de petites quantités d'oxydes métalliques, tels que l'oxyde de magnésium (MgO), l'oxyde de calcium (CaO) ou l'oxyde d'yttrium (Y₂O₃). Cependant, la forme tétragonale est en fait « métastable » à température ambiante. Les contraintes externes telles que le sablage, le meulage, l'impact et le vieillissement thermique

peuvent déclencher la transformation de la phase tétragonale en phase monoclinique ($t \rightarrow m$). Cette transformation est associée à une expansion de volume de 3 à 4 % qui induit des contraintes de compression, fermant ainsi la pointe de la fissure et empêchant la propagation de celle-ci. Cette caractéristique unique est connue sous le nom de durcissement par transformation et peut expliquer l'augmentation de la résistance à la rupture et de la ténacité des céramiques Y-TZP par rapport à d'autres céramiques dentaires.

La résine composite est un autre matériau qui peut être fabriqué par CFAO. Les avantages tels que la facilité d'usinage, la possibilité de réparation intra-orale, un comportement mécanique suffisant et une bonne esthétique ont rendu les composites à base de résine comparables aux matériaux vitrocéramiques (Rüdiger H. et al. 2018) (13). Pour augmenter les propriétés mécaniques, des matériaux composites hybrides, également appelés nanocéramiques ou réseau céramique infiltré de polymère (PICN), ont été développés et contiennent 80 % en poids de particules de zircone et de silice enchâssées dans une matrice de résine de 20 % en poids (Fasbinder D.J. 2012) (14).

Une alternative à la fabrication CAO/FAO soustractive est la fabrication additive, qui consiste à déposer des couches de matériau pour créer un objet en 3D. Le processus d'impression s'effectue couche par couche, par frittage de la poudre, par dépôt d'un matériau thermoplastique fondu ou par photopolymérisation de la résine (Grzebieluch W. et al. 2021) (15). En 2020, un matériau hybride chargé de céramique (VarseoSmile Crownplus, BEGO, Brême, Allemagne) pour l'impression 3D/la fabrication additive de restaurations dentaires définitives de la couleur des dents a été mis à la disposition de la communauté dentaire (Figure 1).

Il est approuvé pour les restaurations d'une seule dent telles que les couronnes complètes, les inlays, les onlays ou les facettes, et a été - à la connaissance des auteurs - le premier matériau imprimable en 3D disponible approuvé pour les restaurations dentaires définitives. Le matériau est un composite hybride à infiltration céramique composé d'une matrice d'ester méthacrylique avec des charges céramiques et peut être classé comme « céramique à matrice de résine » (Graf T. et al. 2022) (16).

En ce qui concerne les protocoles de collage, le collage sur le pilier de la dent est devenu une procédure de routine pour réaliser un scellement esthétiquement durable en utilisant une combinaison de primer adhésif et/ou de ciment de résine. Il est également considéré comme un protocole essentiel pour le scellement des restaurations tout-céramique, en particulier lorsque leur rétention à la structure dentaire est défavorable (Triwatana, Premwara et al. 2016) (17). Les systèmes adhésifs universels ont été lancés dans un but de simplification technique. Ces systèmes adhésifs sont classés comme « universels » car ils peuvent être utilisés comme adhésifs auto-mordançants, adhésifs de mordançage et de rinçage, ou comme adhésifs auto-mordançants sur la dentine et adhésifs de mordançage et de rinçage sur l'émail (mordançage sélectif de l'émail). En outre, ces adhésifs peuvent être appliqués à la surface de différents matériaux de restauration grâce à la présence de monomères fonctionnels dans leur composition (Passia N. et al. 2015) et (Costa DM. et al. 2017) (18,19). Par exemple, les céramiques dentaires à base de zircone ont une quantité minime ou pas de silice dans leur composition, alors les silanes ne seraient plus un apprêt approprié pour les céramiques dentaires à base de zircone pour le collage avec le ciment de résine. Différents primaires adhésifs ont été étudiés,

tels que le coupleur de zircone, certains silanes et des monomères fonctionnels à base de phosphate. L'utilisation de monomères fonctionnels à base de phosphate tels que le 10-méthacryloyloxy-décyl dihydrogénophosphate (MDP) après le sablage de la surface de la zircone est connu pour favoriser la liaison par réaction chimique entre les groupes hydroxyles du monomère fonctionnel et le Zr-OH sur la surface de la zircone (Triwatana, Premwara et al. 2016) et (Yoshida K. et al. 2006) (17, 20). Dans le cas des céramiques contenant de la silice, la partie inorganique du silane réagit chimiquement avec -SiOH sur la surface de la céramique, tandis que le groupe organique méthacrylate se lie chimiquement aux matériaux à base de résine (Plueddemann E. 1991) (21). Une augmentation de la mouillabilité et une réduction de l'angle de contact ont également été signalées après l'application de silane sur la surface de la céramique (Meng X. et al. 2011) (22). En ce qui concerne les céramiques CAD/CAM à matrice de résine, des traitements de surface spécifiques ont été signalés pour augmenter le μ TBS à la structure de la dent, l'abrasion à l'air suivie d'un adhésif contenant du MDP et l'acide fluorhydrique suivi d'une application de silane semblent être les meilleures stratégies pour optimiser la force d'adhérence des céramiques CAD/CAM à base de résine et du réseau céramique infiltré dans le polymère (PICN), respectivement (Fathy et al. 2022) (23).

La surface intaglio des restaurations indirectes doit être préparée distinctement avant l'application de l'adhésif et de l'agent de scellement. Dans ce contexte, Le mordançage à l'acide fluorhydrique (HF) est considérée comme l'une des méthodes les plus efficaces pour préparer la surface des vitrocéramiques, car elle crée une texture micro-rugueuse qui améliore l'adhérence. Malgré son efficacité, l'HF pose de sérieux problèmes de toxicité et de sécurité, ce qui limite son utilisation clinique. Par

conséquent, des alternatives moins dangereuses, telles que les apprêts céramiques automordançants, ont été étudiées. Par exemple, Monobond Etch & Prime a montré des performances de mordantage variables selon le type de matériau céramique utilisé (El-Damanhoury HM et al., 2020) [24]. Une autre technique couramment appliquée est le sablage, ou micro-abrasion avec des particules d'alumine, qui convient particulièrement à la zircone et aux céramiques à base de résine (Stawarczyk B et al., 2012) [25]. Bien que cette méthode permette de créer une surface propre et rugueuse idéale pour le collage, la pression utilisée pendant le sablage est essentielle. Des pressions de sablage élevées peuvent provoquer des microfissures en surface et sous la surface, ce qui affaiblit la force de cohésion du matériau et augmente la probabilité de défaillance à l'interface de l'adhésif (Yoshihara K et al., 2016) [26].

Comme alternative aux protocoles traditionnels tels que le sablage et le mordantage à l'acide, l'utilisation de lasers a été explorée dans la littérature comme une méthode capable de créer une surface avec une microrugosité améliorée. Des études récentes ont suggéré que les lasers pourraient être une alternative plus sûre, car les aérosols d' Al_2O_3 dans le sablage à l'alumine ont été associés à des effets nocifs sur la santé humaine (Albergardi A.B et al. 2025) (27). Les lasers sont devenus un outil à part entière de la dentisterie moderne, couvrant à la fois les procédures chirurgicales et non chirurgicales. La prosthodontie, une branche spécialisée de la dentisterie, a également adopté les lasers, reconnaissant leur valeur significative par rapport aux techniques traditionnelles. L'intégration des lasers a permis d'améliorer la qualité des soins pour les patients et les dentistes, grâce à leur capacité à réaliser des excisions précises, à favoriser une cicatrisation rapide et à améliorer la réaction des tissus pendant les interventions chirurgicales. Les lasers les plus couramment utilisés

en prosthodontie sont les lasers au dioxyde de carbone, à l'argon et à l'yttrium-aluminium-grenat (YAG). Alors que de nombreuses revues traitent de l'utilisation des lasers en dentisterie générale, la littérature traitant spécifiquement des lasers en prosthodontie reste limitée (Binrayes A., 2024) (28).

Dans une mise à jour sur l'application des lasers en prosthodontie, Binrayes A. développe les divers rôles et les progrès récents des lasers dans ce domaine. Il souligne l'intégration des lasers dans les protocoles de diagnostic, les approches thérapeutiques et la fabrication de restaurations prothétiques. Binrayes souligne en particulier que les lasers à Erbium, fonctionnant à des longueurs d'onde de 2780 nm (Er,Cr:YSGG) et 2940 nm (Er:YAG), sont particulièrement efficaces dans la gestion des procédures relatives aux tissus durs. Ces lasers ciblent principalement la teneur en eau des tissus et présentent une grande affinité pour l'hydroxyapatite, ce qui les rend idéaux pour les procédures telles que la préparation des cavités et l'émailoplastie. En outre, les lasers Erbium sont connus pour éliminer les odeurs et les vibrations généralement associées aux pièces à main dentaires, tout en offrant une approche précise et conservatrice de l'ablation des tissus durs avec un minimum de dommages thermiques.

D'autre part, les lasers CO₂, qui fonctionnent à une longueur d'onde de 10 600 nm dans le spectre infrarouge, sont principalement utilisés pour les procédures sur les tissus mous en prothèse dentaire. Ces lasers sont fortement absorbés par les tissus contenant de l'eau, ce qui les rend bien adaptés à l'élimination précise des tissus mous. Les procédures telles que les gingivectomies, les frénectomies, les biopsies excisionnelles et l'allongement des couronnes bénéficient des lasers CO₂. Ils sont

également efficaces pour la dépigmentation des tissus mous. Les lasers CO₂ présentent l'avantage supplémentaire de pouvoir couper et coaguler efficacement les tissus mous avec une faible profondeur de pénétration (Garg N. et al., 2015) (29). En outre, les progrès de la technologie des lasers CO₂, tels que le développement d'un laser d'une longueur d'onde de 9 300 nm, ont élargi leurs applications à l'élimination des tissus durs dentaires, ce qui a rendu les lasers CO₂ de plus en plus efficaces dans la préparation des tissus durs dentaires (Luk K. et al., 2015) (30).

Les mesures de la force d'adhérence sont couramment utilisées pour évaluer l'efficacité des protocoles adhésifs. Par conséquent, la présente étude visait à étudier l'influence de diverses stratégies de préparation au laser des matériaux CAD/CAM sur le mode de défaillance et la résistance à la microtraction interfaciale de ces matériaux en comparant un nouveau laser CO₂ de 9300nm au laser Er:YAG et à la préparation de surface conventionnelle. L'hypothèse nulle testée était que le traitement au laser produit des modes de défaillance et une résistance à la micro traction comparable au traitement de surface conventionnel pour les matériaux testés, quel que soit le type de substrat, y compris la zircone, le disilicate de lithium, les composites à base de résine haute température et haute pression, et les matériaux à base de résine imprimés en 3D.

INTRODUCTION

Dental caries is reported to be one of the oldest and most common diseases found in humans. It is a prevalent chronic infectious disease resulting from tooth-adherent cariogenic bacteria that metabolize sugars to produce acid, which, over time, demineralizes tooth structure (**Rathee M. et al. 2024**) (1). New concepts in operative dentistry proposed by Fusayama (**Fusayama T. 1980**) (2) opened new treatment strategies by means of advanced knowledge in cariology and introduction of dental adhesives. Restorative principles proposed by GV Black (**Black GV. 1917**) (3) such as “Extension for Prevention”, “Retention Form” and “Resistance Form” have been modified and/or replaced by “Minimal Invasive Dentistry” (**Tyas MJ. 2000**) (4). Minimal invasive (MI) approach of cavity preparation basically aimed to minimize the reduction of tooth substances to preserve maximum sound tooth structure (**Sano H. et al. 2020**) (5). The concept of minimally invasive dentistry within cariology has evolved because we now have a better understanding of the disease. We can intercept in its development (**Truman et al. 2002**) (6) and have the technical possibilities to remove a minimal amount of healthy tooth substance and make smaller fillings (**Tyas MJ. 2000**) (4).

The development of dental materials' science has provided the modern dentist with an extensive armoury of materials for restorative procedures. The use of indirect methods for conceiving restorations gave room to fabrication procedures that can optimize the mechanical and physical properties of restorative materials. The advent of high-performance machinable ceramics and composite materials as well as

improved adhesive materials and luting agents has widened the potential applications for indirect restorative procedures in modern dental practice **(Palin W. et al. 2005)** (7).

Likely, new developments in computer technology have led to the success of dental computer aided design and computer aided manufacturing (CAD/CAM) for the elaboration of restorations. Highly sophisticated chair side and laboratory CAD/CAM systems have been introduced to the dental profession **(Mantri et al. 2010)** (8). CAD/CAM technology uses a subtractive method, i.e grinding from a material block to obtain the desired form, and has been developed to solve 3 challenges. The first challenge was to ensure adequate strength of the restoration, especially for posterior teeth. The second challenge was to create restorations with a natural appearance. The third challenge was to make tooth restoration easier, faster and more accurate **(Davidowitz G. et al. 2011)** (9). Following the development of CAD/CAM technology, new materials have been conceived by the industry such as ceramic, composite and hybrid materials with optimized mechanical and esthetic properties **(Marchesi et al. 2021)** (10).

Lithium disilicate glass ceramic is a new generation of heat-pressure ceramic that shows higher fracture strength and bending resistance than the first generation of ceramics reinforced by leucite. It is composed of two components: silica, which serves as a glass matrix, and lithium oxide (Li_2O), which serves as a flux to decrease the increasing temperature of the glass matrix that is raised approximately to 2000 °C from 1100 °C. The application of a silane coupling agent after ceramics are etched with acids at different concentrations is one of the known and recommended methods to increase the strength of the bond between ceramic and tooth substrate. Etching with hydrofluoric acid selectively dissolves surfaces that contain a glass matrix and

crystals. It also helps to increase the surface energy for the implementation of acidified surface silane agent. As a result, the chemical bond between the ceramics' inorganic matrix and the resin cements' organic matrix is improved (**Özdemir H. et al. 2017**) (11).

Another dental ceramic is yttria-stabilized zirconium oxide polycrystals (Y-TZP), adopted by CAD/CAM technology for posterior and anterior fixed partial dentures. Zirconia blanks can be processed by milling and depending on the system, partially or fully sintered blanks are used (**Karakoca S. et al. 2009**) (12). Zirconia is a polymorphic material that exists in three different phases: monoclinic, tetragonal, and cubic. The monoclinic phase is low temperature stable phase. Pure zirconia is monoclinic at room temperature, and this phase is stable up to 1170 °C. Above that temperature, it transforms into a tetragonal phase. At a temperature of 2370 °C, zirconia transforms into a cubic phase, which is stable only at very high temperatures. The tetragonal form may be stabilized at room temperature by adding small amounts of metallic oxides, such as magnesium oxide (MgO), calcium oxide (CaO), or yttrium oxide (Y₂O₃). However, tetragonal form is in fact “metastable” at room temperature. External stresses such as sandblasting, grinding, impact, and thermal aging can trigger tetragonal to monoclinic (t → m) phase transformation. This transformation is associated with 3–4% volume expansion that induces compressive stresses, thereby closing the crack tip and preventing further crack propagation. This unique characteristic is known as transformation toughening and may explain the increased fracture strength and fracture toughness of Y-TZP ceramics compared with other dental ceramics.

Resin composite is another material that can be manufactured by CAD/CAM. Advantages such as an easier machinability, the possibility of intra-oral repair, a sufficient mechanical behavior and good esthetics have rendered resin composites comparable to glass ceramic materials **(Rüdiger H. et al. 2018)** (13). To increase mechanical properties, hybrid composite materials, also called nanoceramics or polymer infiltrated ceramic network (PICN), have been developed and contain 80 wt% of zirconia and silica particle entrenched in a 20 wt% resin matrix **(Fasbinder D.J. 2012)** (14).

Finally, an alternative to subtractive CAD/CAM manufacturing is additive manufacturing, which is the process of depositing layers of material to create a 3D object. The printing process takes place layer by layer, by sintering the powder, depositing a molten thermoplastic material or by light curing of the resin **(Grzebieluch W. et al. 2021)** (15). In 2020, a ceramic-filled hybrid material (VarseoSmile Crown^{plus}, BEGO, Bremen, Germany) for 3D printing/additive manufacturing of tooth-coloured definitive dental restorations became available to the dental community (Figure 1).



Figure 1. A picture of VarseoSmile Crown^{plus} and how restorations are conceived by additive manufacturing. Source: <https://www.bego.com/3dprinting/materials/varseosmile-crown-plus/sample-request/>

It is approved for single-tooth restorations such as full crowns, inlays, onlays, or veneers, and has been—to the knowledge of the authors—the first available 3D-printable material approved for definitive dental restorations. The material is a ceramic-infiltrated hybrid composite consisting of a methacrylic ester matrix with ceramic fillers and can be classified as a “resin matrix ceramic” (**Graf T. et al. 2022**) (16).

Regarding luting protocols, adhesive bonding to tooth abutment has become a routine procedure to achieve aesthetically durable cementation using a combination of adhesive primer and/or resin cement. It is also considered an essential protocol for luting all-ceramic restorations especially when their retention to tooth structure is unfavourable (**Triwatana, Premwara et al. 2016**) (17). Universal adhesive systems were launched with the aim of technical simplification. These adhesive systems are classified as “universal” because they may be used as self-etch adhesives, etch-and-

rinse adhesives, or as self-etch adhesives on dentin and etch-and-rinse adhesives on enamel (selective enamel etching). In addition, these adhesives can be applied on the surface of different restorative materials thanks to the presence of functional monomers within their composition **(Passia N. et al. 2015)** and **(Costa DM. et al. 2017)** (18,19). For instance, zirconia-based dental ceramics have minute amount or no silica in their compositions, then, silanes would no longer be a suitable primer for zirconia-based dental ceramics for bonding with resin cement. Different adhesive primers have been investigated such as zirconia coupler, some silanes and phosphate-based functional monomers. Using phosphate-based functional monomers such as 10-methacryloyloxy-decyl dihydrogenphosphate (MDP) after sandblasting of zirconia surface is known to favour bonding via chemical reaction between hydroxyl groups from the functional monomer and Zr-OH on the zirconia surface **(Triwatana, Premwara et al. 2016)** and **(Yoshida K. et al. 2006)** (17, 20). In the case of silica-containing ceramics, the inorganic part of silane chemically reacts with –SiOH on the ceramic surface, while the methacrylate organic group chemically bonds to resin-based materials **(Plueddemann E. 1991)** (21). An increase in wettability and reduction in contact angle have also been reported after applying silane on ceramic surface **(Meng X. et al. 2011)** (22). Regarding resin-matrix CAD/CAM ceramics, specific surface treatments have been reported to increase the μ TBS to tooth structure, air abrasion followed by an MDP-containing adhesive and hydrofluoric acid followed by silane application appear to be the best strategies for optimizing the bond strength of CAD/CAM resin-based ceramics and polymer infiltrated ceramic network (PICN), respectively **(Fathy et al. 2022)** (23).

The intaglio surface of indirect restorations need to be distinctly prepared before the application of the adhesive and luting agent. In this context, Hydrofluoric acid (HF)

etching is considered one of the most effective methods for preparing the surface of glass ceramics, as it creates a micro-roughened texture that enhances adhesion. Despite its effectiveness, HF poses serious toxicity and safety concerns, which limit its broader clinical use. Consequently, less hazardous alternatives, such as self-etching ceramic primers, have been investigated. For example, Monobond Etch & Prime has shown variable etching performance depending on the type of ceramic material used (**El-Damanhoury HM et al., 2020**) [24]. Another commonly applied technique is sandblasting, or micro-abrasion with alumina particles, which is particularly suitable for zirconia and resin-based ceramics (**Stawarczyk B et al., 2012**) [25]. Although this method can create a clean, rough surface ideal for bonding, the pressure used during sandblasting is critical. High blasting pressures can cause surface and subsurface microcracks, weakening the material's cohesive strength and increasing the likelihood of failure at the adhesive interface (**Yoshihara K et al., 2016**) [26].

As an alternative to traditional protocols like sandblasting and acid etch, the use of lasers has been explored in the literature as a method capable of creating a surface with enhanced microroughness. Recent studies suggested that lasers could be a safer alternative, as the aerosols from Al₂O₃ in alumina blasting have been associated with harmful effects on human health (**Albergardi A.B et al. 2025**) (27). Lasers have become an integral tool in modern dentistry, spanning both surgical and non-surgical procedures. Prosthodontics, a specialized branch within dentistry, has also adopted lasers, recognizing their significant value alongside traditional techniques. The incorporation of lasers has elevated the standard of care for both patients and dentists, thanks to their ability to perform precise excisions, promote rapid healing, and enhance tissue response during surgeries. The most commonly utilized lasers in prosthodontics

include carbon dioxide, argon, and yttrium-aluminum-garnet (YAG) lasers. While numerous reviews discuss the use of lasers in general dentistry, literature specifically addressing lasers in prosthodontics remains limited (**Binrayes A., 2024**) (28).

In an update on the application of lasers in prosthodontics, Binrayes A. elaborates on the diverse roles and recent advancements of lasers in the field. He emphasizes the integration of lasers into diagnostic protocols, treatment approaches, and the fabrication of prosthetic restorations. Specifically, Binrayes points out that Erbium lasers, operating at wavelengths of 2780 nm (erbium-chromium-doped yttrium-scandium-gallium-garnet (Er,Cr:YSGG)) and 2940 nm (erbium-doped yttrium-aluminum-garnet (Er:YAG)), are particularly effective in managing hard tissue procedures. These lasers primarily target water content in tissues and exhibit a high affinity for hydroxyapatite, which makes them ideal for procedures like cavity preparation and enameloplasty. Moreover, Erbium lasers are known for eliminating the odors and vibrations commonly associated with dental handpieces while providing a precise and conservative approach to dental hard tissue ablation with minimal thermal damage.

On the other hand, CO₂ lasers, which operate at a wavelength of 10,600 nm within the infrared spectrum, are predominantly used for soft tissue procedures in prosthodontics. These lasers are highly absorbed by water-containing tissues, making them well-suited for the precise removal of soft tissue. Procedures such as gingivectomies, frenectomies, excisional biopsies, and crown lengthening benefit from CO₂ lasers. They are also effective in soft tissue depigmentation. CO₂ lasers provide the additional advantage of being able to cut and coagulate soft tissue efficiently with a shallow depth of penetration (**Garg N. et al., 2015**) (29). Furthermore, advancements

in CO₂ laser technology, such as the development of a 9,300-nm wavelength laser, have expanded their applications to dental hard tissue removal, which has made CO₂ lasers increasingly effective in dental hard tissue preparation (**Luk K. et al., 2015**) (30).

Bond strength measurements are commonly used to evaluate the effectiveness of adhesive protocols. Therefore, the present study aimed to investigate the influence of various strategies of laser preparation of CAD/CAM materials on the failure mode and the interfacial micro-tensile bond-strength of these materials by comparing a novel 9300nm CO₂ laser to Er:YAG laser and conventional surface preparation. The null hypothesis tested was that laser treatment yields micro tensile bond strength comparable to conventional surface treatment for the tested materials, regardless of the substrate type, including zirconia, lithium disilicate, high-temperature high-pressure resin-based composites, and 3D-printed resin-based materials.

MATERIALS AND METHODS

The layout of the study is schematized in Figure 2.

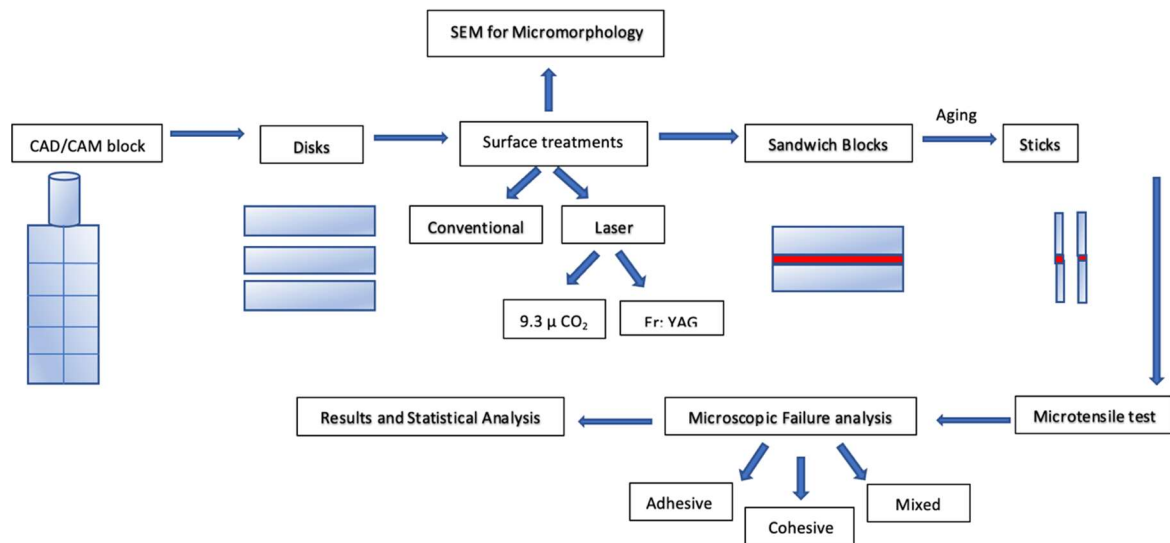


Figure 2. Design of the study from block sectioning to stick preparation for aging and bond strength testing.

Materials selected for this study. Four types of CAD-CAM materials: lithium disilicate ceramic, zirconia ceramic, hybrid ceramic and printed composite were treated by sandblasting or hydrofluoric acid etch as conventional methods, and two different lasers before testing the strength of the bond by microtensile bond strength testing, the luting interface consisting of a universal adhesive containing a silane and 10-MDP monomer and a hybrid composite. The two lasers used in this study are shown in Figures 3 and 4.



Figure 3. An image of the 9.3-micron CO₂ laser: Solea® All-tissue dental laser
<https://www.convergentdental.com/solea>



Figure 4. An image of the Er:YAG laser: LiteTouch ®
<https://orcos.ch/fr/gamme-de-produits/litetouch-iii/>

The CAD-CAM and luting materials used in this study are listed in Tables 1 and 2.

Materials	Brand name	Manufacturer	Composition	Batch number
Zirconia	KATANA Zirconia 14Z L / STML A3	Kuraray Noritake, Tokyo, Japan	ZrO ₂ , Y ₂ O ₃ (~65 wt% cubic content), Zr 60%, O 30%, Hf 1.3%	REF# 125-6612, LOT EFEVA
Lithium-disilicate	IPS e.max CAD	Ivoclar Vivadent; schaan, Leichtenstein	SiO ₂ 60–65%; K ₂ O 15–19%; Al ₂ O ₃ 6–10.5%; other oxides and pigments 0–8%	REF# 686545, LOT Z01270
Hybrid ceramic	KATANA AVENCIA	Kuraray Noritake Dental, Tokyo, Japan	Organic part: UDMA, TEGDMA Inorganic part: 62 wt% aluminum oxide (20 nm), SiO ₂ (40 nm)	REF# 4148- EU, LOT 00985
3D-printed resin-based composite	Varseo Smile Crown Plus	BEGO GmbH Bremen, Germany	Esterification products of 4,40 isopropylidiphenol, ethoxylated and 2-methylprop-2enoic acid, silanized dental glass, methyl benzoylformate, diphenyl(2,4,6-trimethyl-benzoyl) phosphine oxide. Total fillers by weight 30-50%	LOT 600317

Table 1. List of ceramic and composite materials used in this study.

Materials	Brand name	Manufacturer	Composition	Batch number
Universal adhesive	CLEARFIL™ Universal Bond Quick	Kuraray Noritake Dental, Tokyo, Japan	10-Methacryloyloxydecyl dihydrogen phosphate (MDP), Bisphenol A diglycidylmethacrylate (Bis-GMA), 2-Hydroxyethyl methacrylate (HEMA), Hydrophilic amide monomers, Colloidal silica, Silane coupling agent, Sodium fluoride, dl- Camphorquinone, Ethanol, Water.	REF# 3574- EU, LOT 3Q0296
Nanohybrid composite	CLEARFIL MAJESTY™ ES-2	Kuraray Noritake Dental, Tokyo, Japan	Matrix: Bis-GMA, hydrophobic aromatic DMA, and hydrophobic aliphatic DMA, dl- camphorquinone Filler: silanated barium glass (particle size 0.37–1.5 µm) and prepolymerized organic filler. 78 wt%, 40 vol%	REF# 3023- EU, LOT 480073

Table 2. List of materials used for luting purposes.

CAD-CAM blocks consisting on zirconia (KATANA Zirconia), lithium-disilicate glass ceramic (IPS e.max CAD), hybrid ceramic (KATANA AVENCIA) and 3D-printed

resin-based composite (Varseo Smile Crown Plus) were cut into 36 slabs of 4mm width and 5mm thickness (n=9) by using a computer-controlled saw (Struers Accutom-100; Copenhagen, Denmark) at 3500 t/m and 0.05 mm/s. Prior to surface treatment, zirconia slabs were sintered according to the manufacturer's instructions (Programat S1 1600, Ivoclar Vivadent, AG, Schaan, Liechtenstein) at 1550 °C for 120 min, and e.max slabs were crystallized (Programat EP 5010, Ivoclar Vivadent, AG, Schaan, Liechtenstein) at 840 °C for 30 min. The slab surfaces were then standardized using first a 120-grit followed by a 600-grit silicon carbide sandpaper (SiC EXCELLENCE, LAM PLAN SA, GAILLARD, France) for 30 seconds. Before further use the slabs were cleaned in an ultrasonic bath for 5 min (SONOREX, BANDELIN electronic, Berlin, Germany).

Determination of laser parameters for surface treatment. In a pilot test different laser parameters were applied on the materials and examined under an optical microscope at 200x magnification (Keyence VHX5000 Osaka, Japan). Surfaces in which a roughened surface was observed were selected to be used for bond strength testing and laser parameters are detailed in Table 3.

	Conventional surface treatment	CO ₂ Laser surface treatment	Er:YAG Laser surface treatment
Zirconia	Sandblasting with alumina particle size range 27 µm. Blasting pressure range: 0.5-2.5 bar. Duration: Approximately 20 s for an area of 10 mm Distance: Approximately 10 mm Application of universal adhesive system containing silane and MDP.	40% power 100% mist	1.5 W 100 mJ, 15 Hz 3 H ₂ O
IPS e.max	Acid etching for 20 sec. by Hydrofluoric acid (HF) 5% Rinsing by water for 60 sec then cleaning in ultrasonic bath for 5 min. Application of universal adhesive system containing silane and MDP.	40% power 100% mist	4.5 W 300 mJ, 15 Hz 3 H ₂ O
Hybrid ceramic	Sandblasting with alumina particle size range 27 µm. Blasting pressure range: 0.5 to 2.5 bar. Duration: Approximately 20 s for an area of 10 mm Distance: Approximately 10 mm Application of universal adhesive system containing silane and MDP.	10% power 100% mist	1.5 W 100 mJ, 15 Hz 3 H ₂ O
Printed composite	Sandblasting with alumina particle size range 27 µm. Blasting pressure range: 0.5 to 2.5 bar. Duration: Approximately 20 s for an area of 10 mm Distance: Approximately 10 mm Application of universal adhesive system containing silane and MDP.	10% power 100% mist	1.5 W 100 mJ, 15 Hz 3 H ₂ O

Table 3. Detailed preparation steps for each surface.

Micromorphology observation after sandblasting, hydrofluoric acid etch, CO₂ and Er:YAG laser. The slabs corresponding to each of the four materials were prepared following manufacturer's instructions. For example, lithium disilicate ceramic was prepared by 5% hydrofluoric acid etch (conventional method), with the 9.3 µm CO₂ laser (Solea) and with the Er: YAG laser (LiteTouch). The same was performed with the other three materials, zirconia, hybrid ceramic and printed composite. Surface patterns were then further analyzed under a Scanning Electron Microscope (Sigma 300 VP, Karl ZEISS, Cambridge, UK). To this purpose they were mounted on aluminum stubs (stub silver RHP 14054 Baltic preparation E.K., Wetter, (Ruhr) Germany) with a glue (Cementit universal, Wabern, Switzerland) and sputter-coated with gold-palladium of 100 nm by means of a sputter-coater (HHV BT150, West Sussex, United Kingdom). Surfaces were consequently observed under magnifications of 350x, 1000x, 5000x and 10000x and micrographs were procured from each preparation method.

Sample preparation and aging by thermocycling. After specific surface treatments, two identically pretreated slabs of each material were airdried to start with the adhesive protocol. A universal adhesive (CLEARFIL™ Universal Bond Quick, Kuraray Noritake Dental, Tokyo, Japan) was applied on both surfaces and airdried for 5 sec. Then, a preheated nanohybrid composite (CLEARFIL MAJESTY™ ES-2, Kuraray Noritake Dental, Tokyo, Japan) was applied on the prepared surfaces and luted together under a standardized manual pressure of 1Kg. The resulted sandwich slabs were fixed with an initial light polymerization of 2x 20s before manual pressure was removed and continued with 4x 20s from different sides using a light curing unit with a power output of 1200 mW/cm² (FUSION, DentLight, Plano, Texas, USA).

After water storage at 37°C for one week the luted slabs were subjected to a thermocycling test consisting of 5500 cycles between 5°C and 55°C in a thermocycler (TC-4 SD Mecxhatronik, Feldkirchen-Westerham, Germany). The sandwich slabs made out from the four tested materials were then sectioned along the X- and Z-dimension starting from the center with a diamond blade (Struers 150 mm dia X 0.5mm X 12.7mm dia) in order to create 10 serial sticks with a 1mm² squared section and a length of 10mm. The outer part of the sandwich slabs was discarded.

Microtensile bond strength test. After measuring the cross-sectional area of each stick with a digital caliper (VOGEL Kevelaer, Germany) to ensure the 1mm² squared section, specimens were fixed in a universal testing machine. The adhesive interface was protected with a teflon tape (3M GmbH, Neuss, Germany); both ends were then luted to the testing device with a universal adhesive (CLEARFIL™ Universal Bond Quick) and flowable composite (Tetric EvoFlow, Ivoclar Vivadent; Schaan, Leichtenstein). For micro-tensile testing, the fixed sticks were pulled apart with the

testing machine at a crosshead speed of 1 mm/min, a pre-load of 0.5N and a constantly increasing load until fracture. The μ TBS in (expressed as megapascal MPa) was calculated by dividing the load (N) over the measured respective cross-sectional area (mm²) of each stick.

Assessment of the failure mode. Fractured surfaces of each stick were evaluated under a stereomicroscope (LEICA A60, magnification 20X, Leica Microsystems; Switzerland) to determine the fracture modes; they were categorized into four patterns (Figure 5):

- 1) Pretest failure with a fracture of the sample before bond strength testing,
- 2) Adhesive, with a fracture at the luting material interface/stick interface (Figure 5 a),
- 3) Cohesive, where the fracture occurred within the CAD/CAM stick material (Figure 5 b),
- 4) Mixed with the fracture at the luting material / stick interface accompanied by fractures within the part of the CAD/CAM material (Figure 5 c).

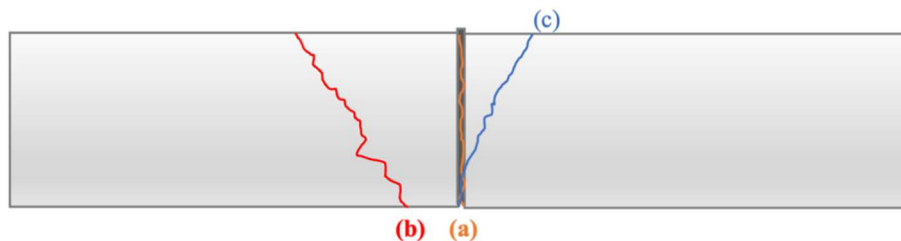


Figure 5. Representation of the different failure patterns observed after the microtensile bond strength test.

Statistical analysis. Normality of data was assessed with Shapiro-Wilk test. Non parametric Kruskal-Wallis and post-hoc test assessed the differences between materials. The level of confidence was set to 95% (SPSS).

RESULTS

The bond strength values of the different groups are detailed in Table 4 and Figure 6.

Groups / surface treatment	Median (MPa)	Post-hoc test
e.max / 5% hydrofluoric acid	28	B,C
Zirconia / sandblasting	36	A,B
Avencia HC / sandblasting	46	A
Avencia HC / Er:YAG laser	26	C,D
Avencia HC / CO ₂ laser	41	A,B
Printed composite / sandblasting	25	C,D
Printed composite / Er:YAG laser	21	D
Printed composite / CO ₂ laser	23	D

Table 4.

Results of the microtensile bond strength test after thermocycling expressed as Median. Groups connected by the same letter are statistically similar at the $p=0.05$ level and apply to the column.

Abbreviations: **Avencia HC**: Avencia hybrid ceramic.

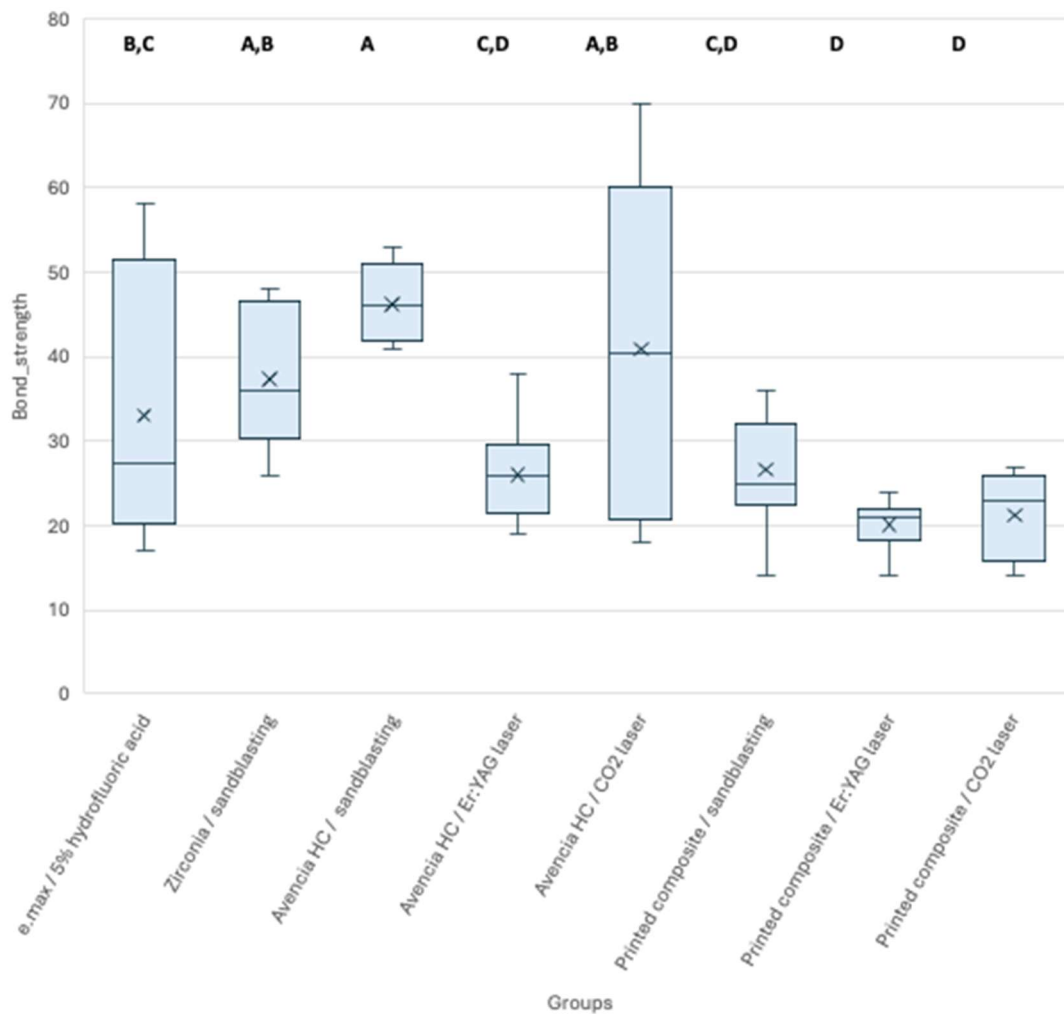


Figure 6.

As a complementary information to Table 4, the box plots of the bond strength results (MPa). Abbreviations: **Avencia HC**: Avencia hybrid ceramic.

The hybrid ceramic (Avencia HC) prepared by conventional method (sandblasting) showed the highest bond strength (Median: 46 MPa), followed by zirconia and lithium disilicate ceramic (e.max) prepared by sandblasting and hydrofluoric acid, attaining bond strengths of 36 MPa and 28 MPa, respectively. CO₂ laser prepared surfaces of hybrid ceramic (Avencia HC) showed comparable bond strength of 41 MPa but with a highly spread data (Figure 6).

Mode of failure expressed as number of sticks for each material and surface preparation method is detailed in Table 5.

Groups	Adhesive failure	Cohesive failure	Mixed failure
e.max / 5% hydrofluoric acid	4	0	6
e.max / Er:YAG laser	PTF	PTF	PTF
e.max / CO ₂ laser	PTF	PTF	PTF
Zirconia / sandblasting	1	2	5
Zirconia / Er:YAG laser	PTF	PTF	PTF
Zirconia / CO ₂ laser	PTF	PTF	PTF
Avencia HC / sandblasting	3	4	1
Avencia HC / Er:YAG laser	7	2	0
Avencia HC / CO ₂ laser	3	4	0
Printed composite / sandblasting	1	8	0
Printed composite / Er:YAG laser	0	6	0
Printed composite/ CO ₂ laser	0	8	0

Table 5.

Failure modes of the four materials after the microtensile bond strength test. Abbreviations: **Avencia HC**: Avencia hybrid ceramic; **PTF**: pretest failure.

Lithium disilicate ceramic surface prepared by 5% hydrofluoric acid etch revealed similar percentage of adhesive (4) and mixed (6) failures, whereas zirconia prepared by sandblasting showed an increased amount of mixed failures (5). The hybrid ceramic (Avencia HC) prepared by sandblasting presented comparable amounts of adhesive (3) and cohesive (4) failures. Printed composite majorly failed due to cohesive failure within the composite (8), not involving the adhesive interface.

Both ceramics, e.max and zirconia, did not effectively react to Er:YAG laser as bond strength tests could not even be performed due to pretest failures. Adhesive failures (7) prevailed in the hybrid ceramic Avencia HC, while the printed composite majorly failed due to cohesive failure within the composite (6), not involving the adhesive interface.

Both ceramics, e.max and zirconia, did not effectively react to CO₂ laser as bond strength tests could not even be performed due to pretest failures. The hybrid ceramic surface (Avencia HC) prepared by CO₂ laser presented similar amounts of adhesive (3) and cohesive (4) failures to failures observed in surfaces prepared by sandblasting. The printed composite majorly failed due to cohesive failure within the composite (8), not involving the adhesive interface.

Micromorphology assessment

The surface micromorphology observed under Scanning Electron Microscopy of zirconia, lithium disilicate, hybrid ceramic and printed composite surface after the different conditioning methods are shown in Figures 7 to 10. Surfaces before any treatment are shown in letter A of Figures 7 to 10. An interesting observation was that before any surface treatment, lithium disilicate surfaces showed irregularities with bubble like formations underneath superficial surface layers (Figure 8 A). Conventional surface methods like hydrofluoric acid etch and sandblasting resulted in an increased surface roughness for all materials, as observed in letter B of Figures 7 to 10.

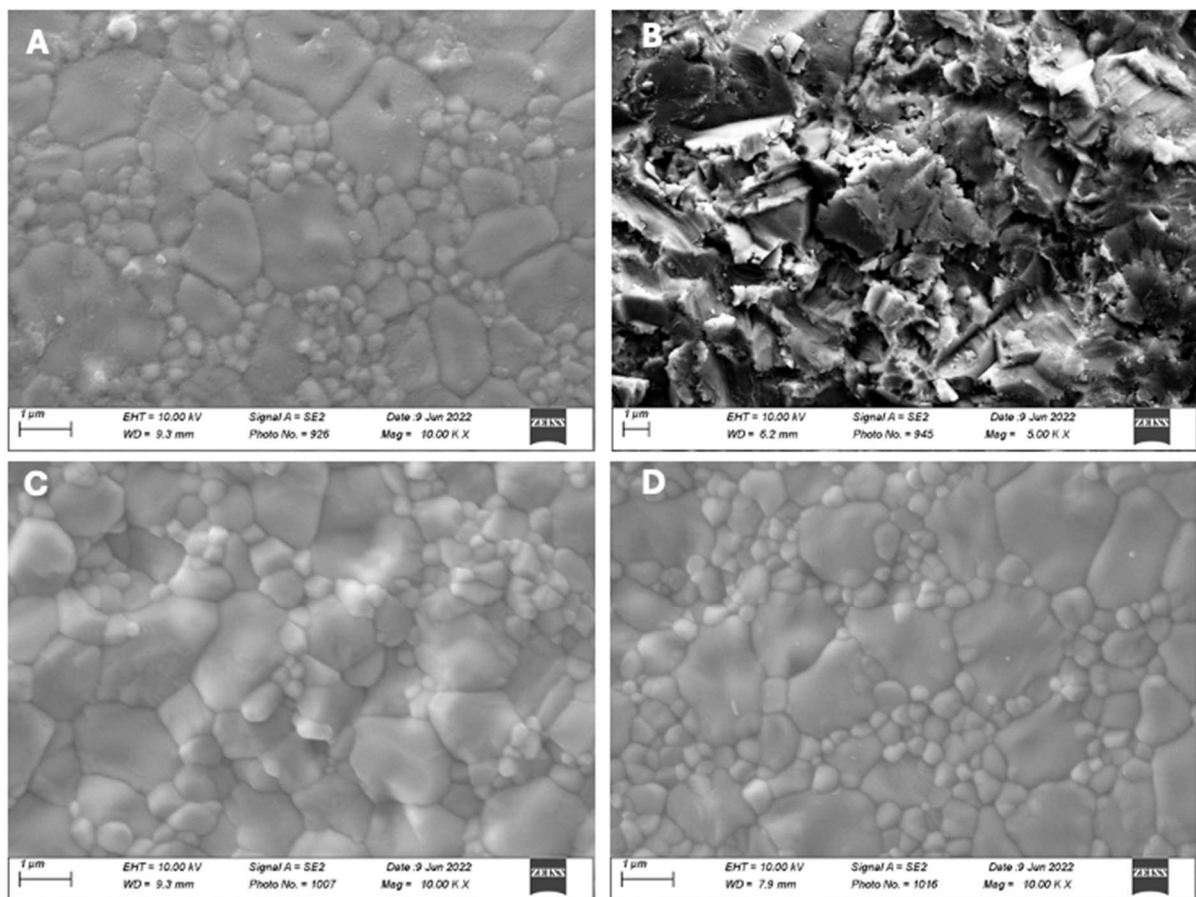


Figure 7. Micromorphology of **Katana Zirconia**. Representative SEM micrographs of A. surface with no preparation, B. after sandblasting, C: after Er:YAG, D: after CO₂.

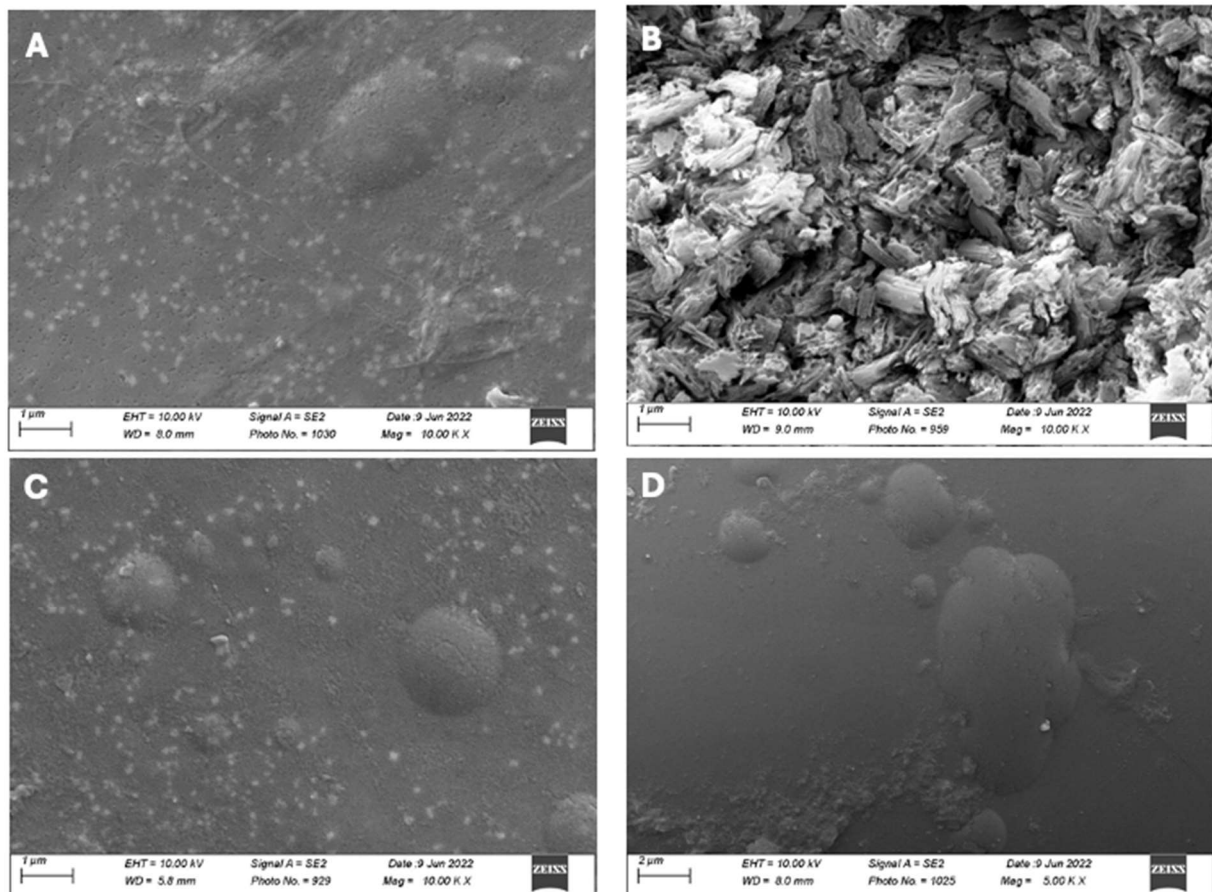


Figure 8. Micromorphology of **lithium disilicate e.max CAD**. Representative SEM micrographs of A. surface with no preparation, B. after 5% hydrofluoric acid etch, C: after Er:YAG, D: after CO₂.

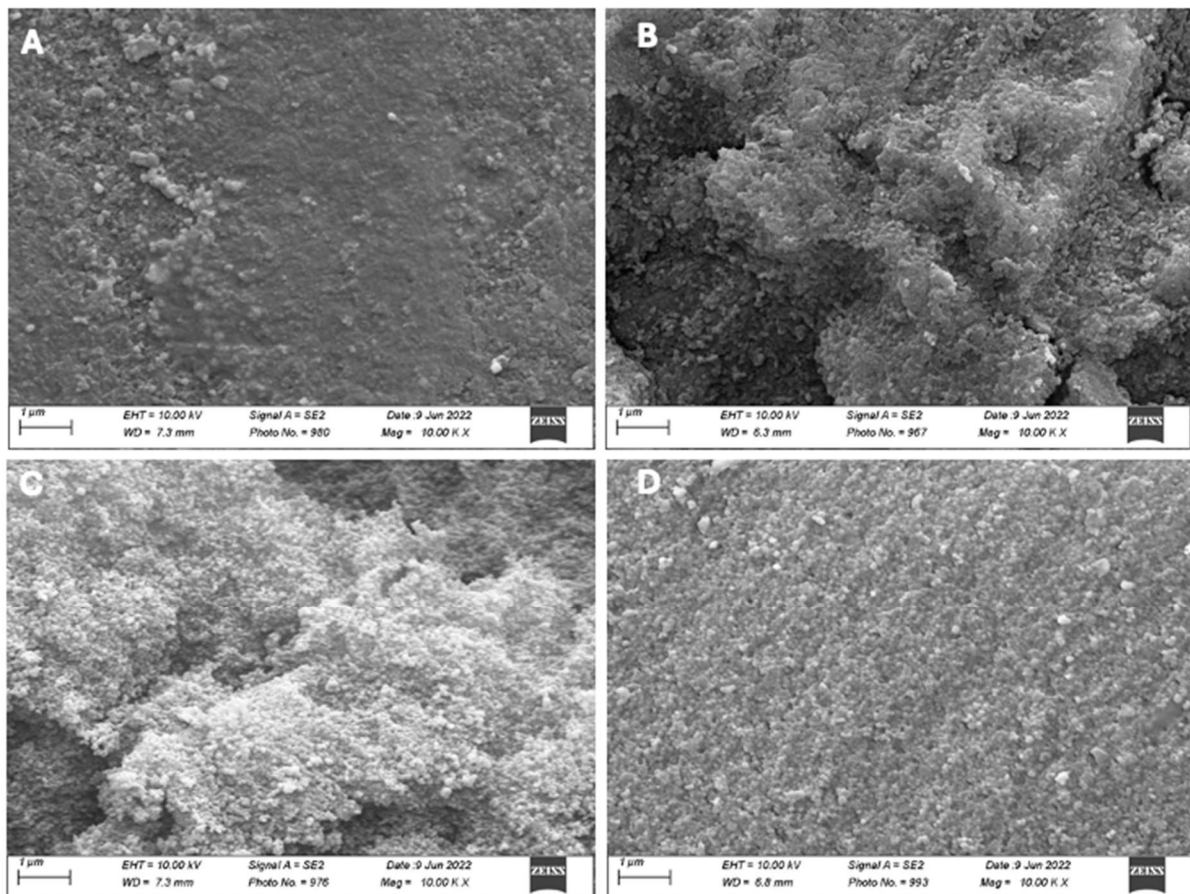


Figure 9. Micromorphology of **hybrid ceramic Katana Avencia**. Representative SEM micrographs of A. surface with no preparation, B. after sandblasting with Al_2O_3 27 microns, C: after Er:YAG, D: after CO_2

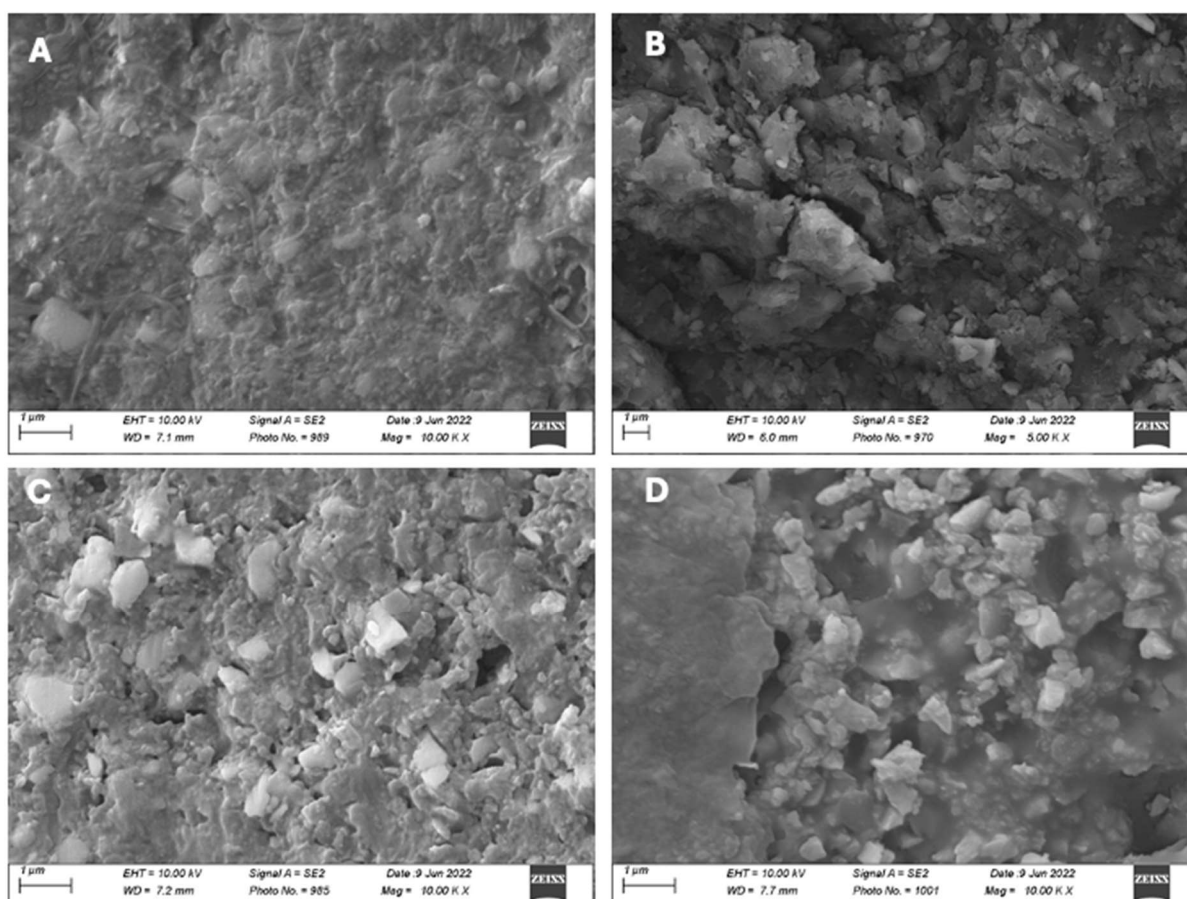


Figure 10. Micromorphology of **printed composite Varseo Smile Crown Plus**. Representative SEM micrographs of A. surface with no preparation, B. after sandblasting with Al_2O_3 27 microns, C: after Er:YAG, D: after CO_2 .

Er:YAG laser-treated surfaces are shown in letter C of Figures 7 to 10. Zirconia surfaces revealed no laser effect (Fig 7 letter C). As for lithium disilicate, no laser effect was either observed (Figure 8 C) as the surface was very similar to the untreated surface (Figure 8 A) with a particular morphology due to the presence of bubbles. Hybrid Ceramic (Avencia HC) presented a rough surface after laser treatment (Fig 9 C) which was quite similar to the one observed after sandblasting (Figure 9 B). Surfaces of printed composite presented a certain roughness after laser treatment (Fig 10 letter C).

CO₂ laser-treated surfaces are shown in letter D of Figures 7 to 10. Zirconia surfaces revealed no laser effect (Fig 7 D) as the surface appeared intact. On lithium disilicate, the effect of laser resulted in a increased size of the bubbles (Fig 8 D) that were already present on the surface with no preparation (Fig 8 A). The effect of CO₂ laser on hybrid ceramic was very similar to the micromorphology after sandblasting, as shown in letters D and B, respectively, of Figure 9. Surfaces of printed composite presented melted areas (Fig 10 letter D).

DISCUSSION

The purpose of this study was to evaluate the surface micromorphology of four CAD/CAM materials prepared with a novel 9.3 mm CO₂ laser, an Er:YAG laser and conventional surface conditioning methods (acid etch and sandblasting). The adhesive interface consisted of two surfaces of the same material conditioned by laser or sandblasting or acid etch, depending on the CAD/CAM material, and then bonded with a universal adhesive (Clearfil Universal Bond) and a nanohybrid resin composite (Clearfil Majesty). A microtensile bond strength test was performed after thermocycle aging. Based on the results, the null hypothesis stating no difference in microtensile bond strength for the tested surface conditioning methods had to be rejected; differences were observed between materials depending on each preparation method.

The methodology. A microtensile bond strength test was chosen to assess the reliability of the bonding interface in materials commonly used for indirect restorations. Testing the bond strength by tensile loading produces more adhesive failures which may be of clinical relevance as adhesive failures can cause the detachment of restorations (**Della Bona A. et al, 1995**) (31). Shear bond strength could have been also used to test the strength of the bond. Meanwhile, the failure mode occurring after shear bond testing is often cohesive within the substrate rather than adhesive at the interface (**Della Bona A. et al, 1994 - Chadwick RG. Et al, 1998**) (32-33). In this sense, Sano et al (**Sano et al, 1994**) (34) described that the introduction of the microtensile bond strength test has shifted the failure pattern further to occur at the adhesive interface due to a more uniform stress distribution during loading. Also, differences between shear and tensile tests might be significant when dentin is involved as a bonding substrate (**Scherrer et al. 2010**) (35) but not statistically

different in ceramic-to-ceramic bonding (**Valandro LF. et al. 2008**) (36) justifying why microtensile test was performed in this study.

Fracture analysis and the failure patterns. Adhesive, cohesive and mixed failures observed after the microtensile test were reported and included in the statistical analysis. This was decided after considering several points:

1. It was important to document in a clear and transparent way all study variables and results so as to avoid any *reporting bias*, i.e., when research results are deliberately presented incompletely or inaccurately with the intention of eliminating negative or undesirable results. When this happens results can not be reproduced by the scientific community because they have been distorted during the analysis stages (**van der Steen et al. 2019**) (37).
2. Cohesive failures are often considered a negative outcome and this is a matter of debate. In fact, a previous meta-analysis found that high bond strengths are often associated with cohesive failures (**Leloup et al. 2001**) (38). These authors reported that the higher the bond strength, the higher rate of cohesive failures can be observed. On the contrary, when the bond at the adhesive interface is weak, the failure will grow at the adhesive interface, resulting in an adhesive failure. This is a very important issue because excluding cohesive failures could completely change variable outcomes and their interpretation.
3. Excluding cohesive failures can give misleading results depending on the material. This was the case in a recent study from the group of Professor D.C.Watts (**Franz A. et al. 2022**) (39) and these authors do not recommend exclusion of data based on cohesive fracture behavior.

4. Finally, careful attention must be given to publications recommending the exclusion of cohesive failures from bond strength results and reading them critically is of paramount importance. For instance, three studies recommended the rejection of cohesive failures from bond strength data (**Versluis A. et al. 1997, Cho BH. and Dickens SH. 2004, Ghassemieh E. 2008**) (40,41,42). Meanwhile, the first one used a simulation model and two major factors affecting adhesion were not modelled in their set-up: the anisotropy of dentin and the hybrid layer. The second one evaluated the failed bonded interface with a scanning electron microscope (SEM) without fixing the specimens, the authors even stated that some of the observed cracks/failures could be originated or increased by the effect of vacuum dissection at the SEM chamber. Finally, the third study evaluated the importance of sample shape in bond strength tests, based on a finite element model and not on real samples.

The interest of laser conditioning. Two different types of lasers were chosen. The first one is an Er:YAG laser which emits photons of 2,940 nm wavelength. Thanks to their affinity to water that is present in the tissues, this laser finds a large area of applications in dentistry, on both soft and hard tissues, being the reason why they are called “all-tissue” lasers. The second type is one laser among the CO₂ family of lasers (9,300, 9,600 and 10,600 nm), the surgical laser par (for) excellence, used in dental surgery and experimentally in caries prevention. While the 10,600 nm laser is used mainly in soft tissues, recently a new CO₂ laser at 9,300 nm was proposed for use on both hard and soft tissues (**Olivi et al. 2015**) (43). The main difference between the two laser technologies is that a 9,300 nm carbon dioxide laser has significantly higher absorption coefficients to hydroxyapatite than Er:YAG and other CO₂ laser family (9,600 and

10,600 nm). In addition, it requires less energy to achieve the same absorption effect as other wavelengths. Therefore, it causes much less heat deposit which renders it a desirable tool for application in dental hard tissues (**Xue VW et al. 2021**) (44). Extending their application to indirect restorations may be of clinical relevance, especially when looking for alternatives to acid etch and sandblasting, as will be further discussed.

The use of a universal adhesive as bonding agent. Universal adhesive systems have gained popularity among clinicians seeking for technical simplification. These adhesive systems are classified as “universal” because they may be used as self-etch adhesives, etch-and-rinse adhesives, or as self-etch adhesives on dentin and etch-and-rinse adhesives on enamel (selective enamel etching). In addition, these adhesives can be applied on the surface of different restorative materials thanks to the addition of 10-MDP monomer and silane within their composition (**Costa DM et al. 2017**) (45). These multi-mode generation of adhesives have shown comparable clinical performance to the one of gold-standard etch-and-rinse and self-etching adhesives (**Peumans M et al. 2014**) (46). Therefore, clinically relevant information derived from this in vitro evaluation may be useful to clinicians.

Acid etch and sandblasting for surface conditioning. CAD/CAM materials surface can be treated chemically or mechanically (**Blatz, M.B. et al, 2003 – Strasser T. et al, 2018**) (47,48). The chemical treatment is usually done with 5% or 9 % hydrofluoric acid (**Tian T. et al. 2014**) (49). The acid reacts with silica to form hydrofluorosilicic acid, creating a space for resin cement and resulting in a rougher surface. A silane coupling agent is used after the acid to enhance the chemical adhesion to CAD/CAM

materials. It combines silane, phosphoric, and sulfide methacrylate (**Alex G. 2008 - Alsaeed AY. 2022**) (50-51). The mechanical treatment is done by subjecting the intaglio surface to air particle abrasion with 25, 27, or 50 microns of aluminum oxide. Sandblasting with alumina air particles showed a significant surface alteration in hybrid ceramic and zirconia (**Aboushelib M.N. et al. 2014**) (52). In the case of zirconia, it cannot be roughened with etching gel due to the lack of silica, so the bonding involves three steps: air particle abrasion, zirconia primer, and adhesive resin (**Blatz et al. 2016 – Fasbinder D.J. 2012**) (53-54).

In this study the bond strength for zirconia and LiSi₂ ceramic prepared by sandblasting and acid etch was similar to the values reported by previous studies (**Triwatana P. et al, 2016**) (55).

Interestingly, the spread of the results in LiSi₂ ceramic after acid etch (Figure 6, 1st box plot from the left) could be due to the presence of bubbles on the surface that interfered with the acid effect. In an attempt to explain the mechanism of this finding, **Serbena et al. 2015** (56) reported on the evolution of crystallization of a stoichiometric lithium disilicate glass, from the parent glass up to the fully crystallized material. They observed the formation of bubbles starting at approximately 65% crystallinity and crystal size > 10 µm, which reached 5% of the sample volume for the 100% crystalline glass- ceramic. They demonstrated that these bubbles reduced the mechanical properties of their glass-ceramic. Other few authors as **Akatsuka et al. 2019** (57) have reported on bubble nucleation triggered by crystallization, they analyzed the cause of bubble formation in more depth in a surface crystallized Na₂MnP₂O₇ glass and found that H₂O was the primary source of bubbles. Another interesting case is SiO₂–B₂O₃–P₂O₅, especially along the SiO₂-BPO₄ join, where so-called “gas ceramics” were found, based on bubbles of hydrogen (**Holland W. et al. 2012**) (58). In this case, as

crystalline BPO₄ is nucleated and crystallizes, H₂ bubbles form. To produce these micro foams, the phosphate must be batched with ammonium phosphate, which is the source of hydrogen. It is possible that these hydrogen-containing bubbles over the surface interfered with the etching effect on ceramic surface, creating areas where the acid reached the ceramic and others where it did not. This phenomenon could explain the spreading of the bond strength results.

Lithium disilicate (IPS e.max CAD). No ablative effect of Er:YAG laser was observed on lithium disilicate. This can be explained by the interaction of the laser wavelength with the composition of the ceramics. Absorption bands of silica (peak at 1000 cm⁻¹) which is the main component of LiSi₂ ceramic do not coincide with the maximum absorption of an Er:YAG laser. Confirming our findings, **Özdemir H. et al., 2017** (59) studied the effect of different surface treatments on bond strength of different resin cements to lithium disilicate glass ceramic and they concluded that laser irradiation (Er: YAG) was not an effective method as lithium disilicate does not interact with the laser beam because of its high reflectivity. This can be explained by lithium disilicate composition: SiO₂ (60–65%), K₂O (15–19%) Al₂O₃ (6–10.5%) other oxides and pigments (0–8%). As silica absorption peak is around 1000 cm⁻¹, absence of water absorption bands leads to no Er:YAG laser effect on the material itself.

Zirconia (Katana Zirconia). No ablative effect of Er:YAG laser was observed on zirconia. This can be explained by the interaction of the laser wavelength with the composition of the ceramics. Absorption bands of ZrO (peak at 560 cm⁻¹) which is the main component of zirconia ceramic, do not coincide with the maximum absorption of an Er:YAG laser. Even though previous research showed an increased bond strength after laser surface treatment using different dental lasers including CO₂ laser, Er:YAG

laser, Er,Cr:YSGG laser, and Nd:YAG laser, conflicting results are frequently reported in the literature (**Han J. et al. 2021**) (60). This is because the results of laser on zirconia highly depend on the laser parameters. For instance, with long pulse duration and relatively low laser power, hardly any material is removed by laser ablation. Sometimes there is even a polishing effect with long pulse laser treatments of originally rough surfaces. Another study (**AramiS 2014**) (61) concluded that CO₂ lasers adversely affected the mechanical properties of zirconia due to a large amount of heat and microcracks, as CO₂ laser irradiation causes heat induction and rapid expansion on the surface of the zirconia ceramics, and the subsequent contraction during solidification causes shell-form porosities (**Akın H. et al. 2011- Kasraei S. et al. 2014**) (62-63).

Hybrid ceramic (Katana Avencia). The hybrid ceramic (Avencia HC) prepared by sandblasting attained bond strength scores of 46 MPa; after CO₂ laser the scores were of 41 MPa and these two values were statistically similar (Table 4). Moreover, Figure 9 showed a similar micromorphology whether the surface was sandblasted (Fig 9 B) or CO₂ laser-prepared (Fig 9 D). Possibly, CO₂ laser could had an effect on AVENCIA HC by removing the hydroxyl groups from composites surface, facilitating bonding to this surface (**Hartwig A. et al. 1996**) (64).

Printed composite (Varseo Smile Crown Plus). The failure mode of the printed composite revealed major predominance of cohesive failures within the material, rendering impossible the measurement of bond strength. One point of view could be that a low percentage of SiO₂ fillers (30-50%) together with a high percentage of organic matrix might explain the low resistance of this material against tensile forces.

But another point of view is that cohesive failures occurred due to a highly resistant adhesive interface, making failure to occur cohesively within the material, keeping intact the adhesive interface. This hypothesis warrants further investigation as these printing materials are quite new in the dental market.

Finally, there are some limitations of this study. Bond strength was tested after thermocycling with no mastication forces applied on samples to simulate the intraoral aging process. Also, only one universal adhesive and hybrid composite was used as luting agent, rendering the results only relevant when using the materials tested in this study.

CONCLUSIONS

- Sandblasting zirconia and the hybrid ceramic showed comparable and the highest results of bond strength.
- Treating the surface of the hybrid ceramic by CO₂ laser showed high bond strength results as well, however, the huge spreading of data raises into question the reliability of laser preparation on this surface.
- Laser could not condition the surface of zirconia and lithium disilicate, rendering impossible the bond strength test.
- In respect to printed composites, cohesive failures within the material prevailed over adhesive ones, regardless of the surface preparation method.

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